Peristaltic Blood Pump Design Workflow

4-Stage Pipeline with Required Input Data

This part of project focuses on the optimization of a peristaltic blood pump designed for hemodialysis extracorporeal circuits. The primary objective of the optimization is to minimize hemodynamic trauma by suppressing transient high-pressure pulses and reducing fluid shear stress during the cyclic compression and decompression of the Tygon tube. By integrating analytical kinematics of peristaltic motion with multi-physics numerical simulations, the pump's geometry and operational dynamics were refined to reliably overcome circuit pressure losses while ensuring smooth, continuous, and biocompatible blood transport.

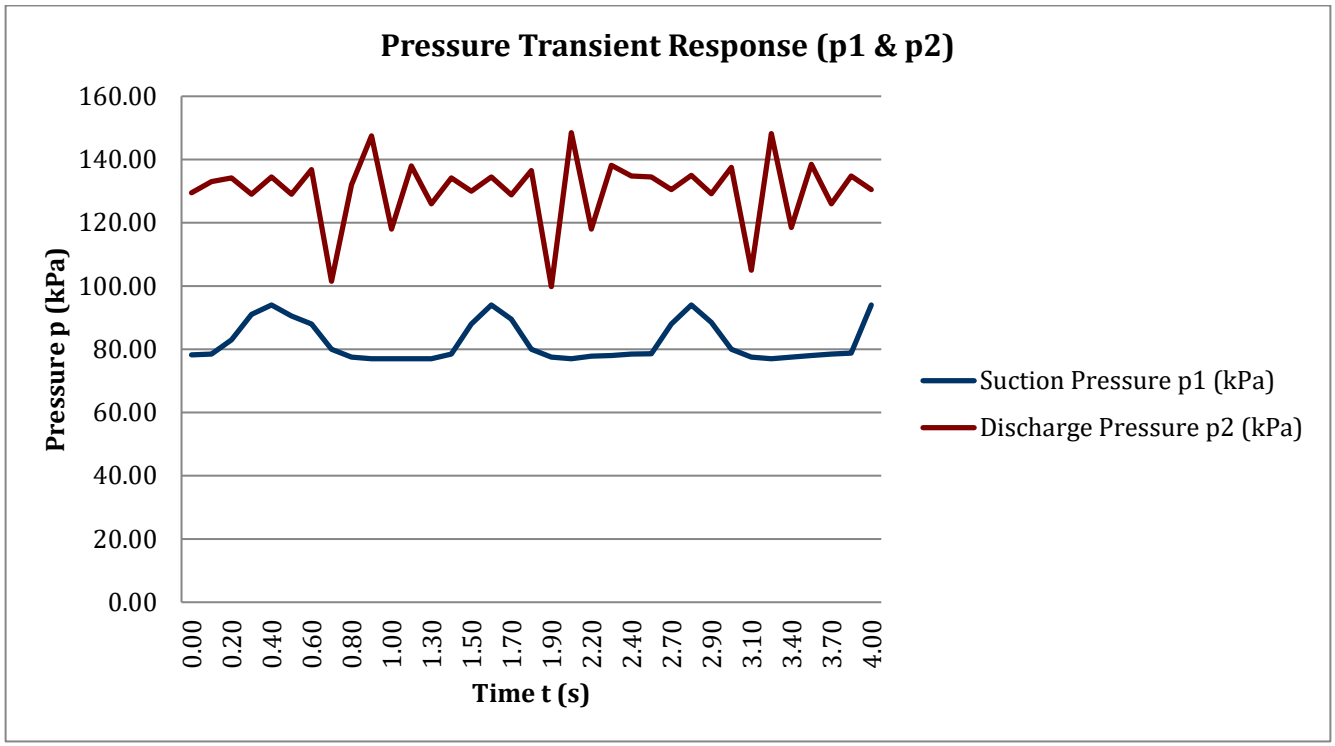


Figure1: Time-Series Pressure Data

The core objective of this work is the structural and kinematic optimization of a rotary peristaltic blood pump to ensure gentle and efficient extracorporeal blood circulation.

Key Optimization Objectives:

i. **Pressure Pulse Attenuation:** Suppress high-amplitude pressure spikes and backflow reflux during the sudden tube compression and decompression phases.
ii. **Shear Stress & Hemolysis Reduction:** Minimize mechanical shear stress levels imposed on erythrocytes to prevent blood trauma and cell damage.

iii. **Pressure Loss Compensation:** Deliver sufficient head pressure to overcome total dynamic pressure drops across the arterial access, dialyzer filter, and venous return lines.

iv. **Kinematic & Structural Refinement:** Synthesize analytical derivations of peristaltic movement with FEA/CFD co-simulations to optimize casing contours, roller engagement angles, and Tygon tube deformation.

Stage 1: Requirements, Physiological Sizing & Analytical Equations

Parameter	Engineering Inputs & Values	Governing Equation	Primary Design Focus
Flow Rate & System Pressures	• Target Flow (Q): 350 mL/min (5.833×10^{-6} m^3/s) • Suction Pressure (p_1): -13,332 Pa (-100 mmHg) • Venous Return Pressure (p_3): 10,665 Pa (80 mmHg) • Dialyzer Loss (p_dialyzer): 16,665 Pa (125 mmHg) • Discharge Pressure (p_2): 27,331 Pa (205 mmHg) • Differential Pressure (Δp): 40,663 Pa	$p_2 = p_3 + p_{dialyzer} = 27331$ Pa $\Delta p = p_2 - p_1 = 40663$ Pa	Define pressure head requirements across extracorporeal hemodialysis circuit.
Specific Work & Power Output	• Fluid: Blood (ρ = 1050 kg/m^3, ν = 2.65 mm^2/s) • Estimated Efficiency (η_est): 30% • Specific Head (ΔY): 38.73 J/kg • Hydraulic Power Output (P_h): 0.79 W	$\Delta Y = \frac{p_2 - p_1}{\rho} = \frac{27331 - (-13332)}{1050} = 38{,}73 \frac{J}{kg}$ $P_h = \frac{Q \cdot \Delta Y \cdot \rho}{\eta_{est}} = \frac{\frac{350}{60 \cdot 10^6} \cdot 38{,}73 \cdot 1050}{0{,}3} = 0{,}79$ W	Drive sizing estimation for department motor inventory.
Geometric Sizing & Rotational Speed	• Casing Diameter (D): 160 mm • Roller Diameter (d_roller): 30 mm • Number of Rollers (N_roller): 2 • Tube Inner Diameter (d_tube): 13 mm • Wall Thickness (t): 1.5 mm | Occlusion (Oc): 20% • Occlusion Gap (s): 2.4 mm | Outer Dia (D_tube): 16 mm • Rotor Radius (R_r): 62.5 mm | Length (l): 226.2 mm • Volume (V_g): 17.58 mL | Speed (n): 6.24 RPM	$Q_{th} = V_g \cdot n \cdot N_{roller}$ $n = \frac{Q_{th}}{\left(\pi \frac{R + r}{2} - d_{roller}\right) \cdot \left(\frac{\pi d_{tube}^2}{4}\right) \cdot N_{roller}}$ $s = 2t(1 - Oc) = 2.4$ mm	Establish kinematic dimensions and rotational speed.

Stage 2: Material Selection & Hyperelastic Structural Analysis

Step	Technical Specification & Input Values	Governing Equation	Primary Design Focus
Hyperelastic Material Formulation	• Material: Flexible Tygon Tubing • Non-linear strain energy potential model • Experimental Young's Modulus (E): 1.3 MPa • Experimental Shear Modulus (G): 4.0 MPa • Poisson's Ratio (μ): ~0.49 (nearly incompressible)	$W = \frac{1}{2}G(\lambda_1^2 + \lambda_2^2 + \lambda_3^2 - 3)$ $= \frac{1}{2}G(I_1 - 3)$ $G = NkT; \quad G = \frac{E}{2(1+\mu)}$	Accurate stress-strain response modeling under hyperelastic compression.
Boundary Conditions & Mesh Setup	• Fixed support on tube inlet/outlet and casing interface • Displacement boundary condition driven by 30 mm roller motion • Occlusion gap preset to 2.4 mm (20% squeeze)	Displacement Profile: $u_r(t) = f(\theta_{rotor}(t))$ Fixed rim: $u_x = u_y = u_z = 0$	Simulate continuous cyclic tube squeezing without mesh distortion.
Structural Outputs & Life Prediction	• Maximum Von-Mises Stress distribution • Reaction squeezing force on rotor arm • Fatigue life prediction under cyclic deformation	$N_f = C \cdot (\Delta\sigma)^{-\beta}$ $S_{ij} = \frac{\partial W}{\partial E_{ij}}$	Prevent tube spallation, rupture, and excessive shaft deflection.

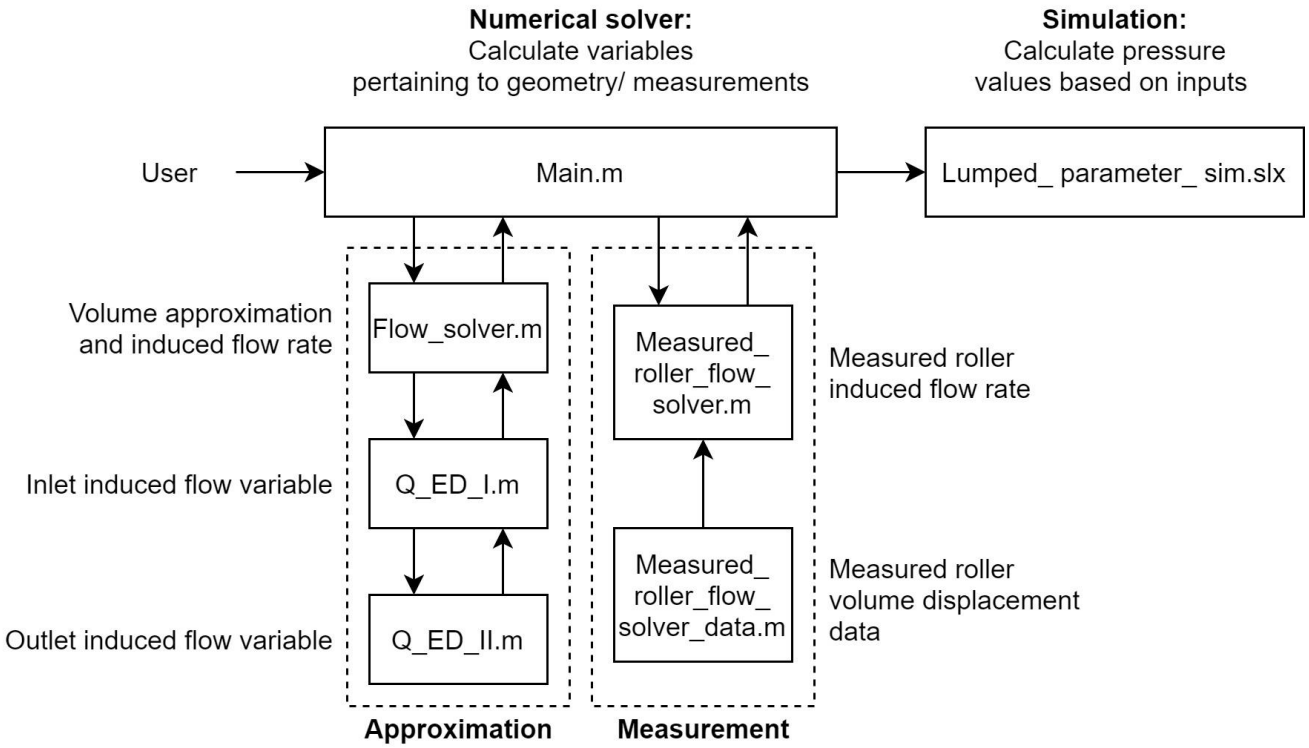
Stage 3: Fluid Dynamics, Shear Stress & Hemolysis Minimization

Parameter	Numerical Setup & Biological Criteria	Governing Equation	Primary Design Focus
Fluid Properties & Navier-Stokes Setup	• Medium: Human Blood • Density (ρ): 1050 kg/m^3 • Kinematic Viscosity (ν): 2.65 mm^2/s • Dynamic Viscosity (μ): ~0.00278 Pa·s • Dynamic / Moving Mesh UDF in ANSYS Fluent	$\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla\vec{v}\right) = -\nabla P + \mu\nabla^2\vec{v} + \vec{f}$ $\nabla \cdot \vec{v} = 0$	Resolve transient laminar flow field inside moving tube cavity.
Shear Trauma & Pressure Pulse Evaluation	• Evaluation of 4 fluid-mechanical damage mechanisms: 1. Velocity gradients (shear between layers) 2. Pressure variations (cell deformation) 3. Flow stagnation (thrombosis risk) 4. Reverse reflux during	$\tau = \mu\frac{\partial u}{\partial y}$ $\Delta P_{reflux} = \xi \cdot \frac{\rho v^2}{2}$	Minimize blood trauma and erythrocyte hemolysis.

	disengagement		
Elliptic Casing Trajectory Optimization	• Elliptic casing variant to accelerate rollers • Reduces pressure gap in front/behind leading roller • Lower pressure pulse amplitude during tube restitution	$r(\theta) = \frac{ab}{\sqrt{(b\cos\theta)^2 + (a\sin\theta)^2}}$ Pressure Gap $\rightarrow \min(\Delta P_{pulse})$	Eliminate backflow pressure spikes and improve flow smoothness.

Stage 4: System-Level Dynamic Simulation

Subsystem	Model Component & Parameter Setup	Governing Equation	Co-Simulation Focus
Electric Motor Drive Subsystem	• DC Motor / BLDC Drive model • Torque Constant (K_t), Back-EMF Constant (K_e) • Resistance (R), Inductance (L), Rotor Inertia (J) • Closed-loop PI speed controller	$T_m = \quad_t \backslash cdot$ $V = IR + L\frac{dI}{dt} + K_e$	Simulate motor electrical dynamics and shaft speed regulation.
Rotor Kinematics & Roller Interaction	• Converts angular velocity (ω) to roller displacement • Models roller-tube contact kinematics across 270° trajectory • Integrates support roller reaction forces	$\theta(t) = \int \omega(t)dt$ $x_{roller}(t) = R_r \cdot \cos(\theta(t))$	Translate rotational motion to cyclic tube squeezing.
Hyperelastic Tube Squeezing Torque Block	• Non-linear deformation resistance torque • Integrates FEA lookup table for reaction force vs compression angle	$T_{load}(\theta) = F_{squeeze}(\theta) \cdot R_r + T_{friction}$	Provide realistic mechanical loading to electric motor.
Hydraulic Circuit & Pulsation Output	• Instantaneous mass flow rate m_dot(t) • Suction reservoir (p_1 = -13,332 Pa) • Dialyzer pressure drop model (p_dialyzer = 16,665 Pa) • Venous output line monitoring	$\dot{m}(t) = \rho \cdot A(t) \cdot \sqrt{\frac{2\Delta P(t)}{\rho}}$ $Q_{out}(t) = \int \quad \dot{m}(t)dt$	Evaluate dynamic flow ripple, pressure pulsation, and pump delivery.


Numerical solver:
Calculate variables
pertaining to geometry/ measurements
Simulation:
Calculate pressure
values based on inputs
User
Main.m
Lumped_ parameter_ sim.slx
Volume approximation
and induced flow rate
Flow_solver.m
Measured_
roller_flow_
solver.m
Measured roller
induced flow rate
Inlet induced flow variable
Q_ED_I.m
Measured_
roller_flow_
solver_data.m
Measured roller
volume displacement
data
Outlet induced flow variable
Q_ED_II.m
Approximation
Measurement